RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 1

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Reg No : 22TD0053

Subject Name : ENTERPRISE SOLUTIONS

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Unit : I

Teacher's Sign :

Department Vision and Mission.

Vision:

Cultivate student as technocrat, researchers and entrepreneurs in the field of computer science and engineering.

Mission:

* To impart quality education in computer science and engineering by means of learning techniques and value-added courses.

* To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.

* To build leadership skills, ethical values and teamwork among the students.

* To strengthen the collaboration of department and industry through internships, mentorships, and professional body activities.

Enterprise System Architecture

An enterprise resource planning (ERP) system is the software responsible for automating the core business processes within a particular company and optimizing their management to achieve the highest performance.

The mordern ERP systems are a set of modules, each of which is involved in the management and automation of a specific business process. procurement, sales, production, accounting and tax accounting, personal records, customer support, CRM, warehouse, logistics, etc. Thus, their architecture determines how modules are interconnected.

In the process of development, the architecture of ERP systems was significantly improved. Four types of architecture emerged in succession. The first type is a two tier architecture, where the system consist of two parts the user and the server side. The second type is complicated by the presence of a third level, this approach excludes direct interaction between the client and the ERP system database, which reduces the performance requirements of user devices. The third type is the web architecture in which the presentation layer is divided into web services and web browsers. The fourth type is a service-oriented architecture or SOA, the size of which grows in direct proportion to the number of service that a particular ERP system provides

Types of ERP Architecture

* Monolithic system.
* Post modern system.

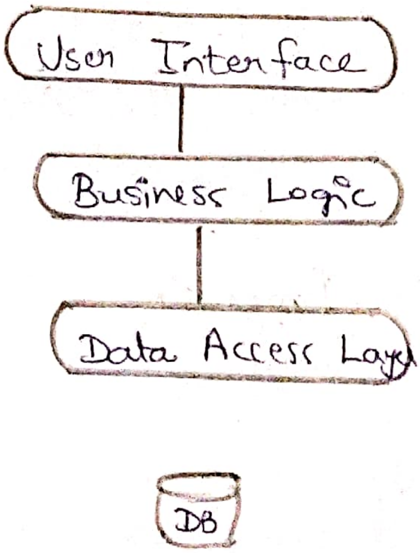
Monolithic System

A monolithic ERP system is characterized by a unified structure, typically developed by a single software vendor. For users, interaction with these system is straightforward. The monolithic structure carriers certain limitations, primarily in flexibility and adaptability.

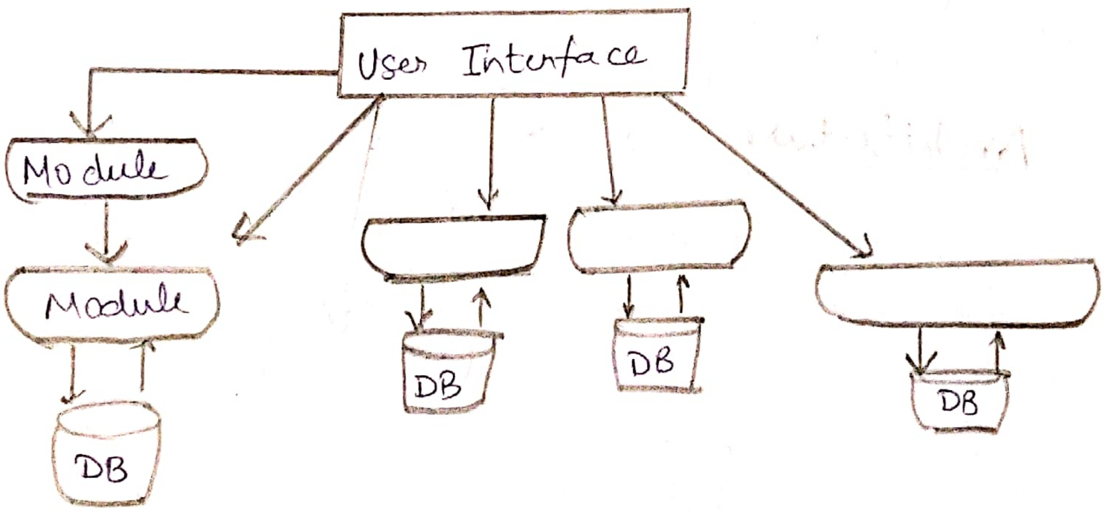
Post moden System.

Post moden ERP system present a more adaptable architecture. They are modular, meaning they are made up of several independent services that can be integrated or removed as needed. This provides an ease of upgradability not typically found in monolithic systems structures. Enterprises can add in ERP features they actually ERPs with features that may b dormant. This system comes with its own set of challenges upgrading these systems can be a complex task. as each module, along with the core functionality, needs to be individually updated.

Monolithic ERP Architecture



Postmodern ERP Architecture:



Key features

* Scalability - Handles increasing data and users
* Flexibility - Adapts to business growth
* Interoperability - Connects with external tools
* Security - protects sensitive business data.
* Automation - Reduces manual workload through AI and workflows.

The ERP system architecture provides technical functionality, user interface, and platform support - It should be able to absorb new technologies.

2. Online Analytical Processing (OLAP)

Online Analytical processing (OLAP) is a computing approach used for multi dimensional data analysis in business intelligence (BI) and decision making. It enables fast retrival of complex queries, helping organisations analyze large datasets efficiently,

OLAP Architecture

OLAP systems consist of multiple layers that facilitate fast and efficient data analysis.

a) Data source Layer.

* Contains raw data collected from databases, ERP, CRM, IOT devices and external sources.
* Extracted using ETL (Extract, Transform, Load) processes to prepare data for analysis.

b) Data ware house layer:

* Stores Integrated, historical data optimized for analysis.
* Uses schemas like star schema and snowflake like schema to organize data efficiently.

c) OLAP Engine

* Converts relational data into multidimensional OLAP cubes.

* Performs aggregation, calculations and indexing to optimize queries.

d) Presentation Layer.

* Provides dashboards, reports and visualization tools for business users.

* Supports interaction through BI tools like power BI, tableau, SAP Business Objects, and Excel pivot tables.

Types of OLAP systems

1) MOLAP (Multidimensional OLAP)

* Stores data in multidimensional OLAP cubes instead of relational tables.
* Fastest query performance due to precomputed aggregates.
* Used for complex calculations and rapid analysis.

2) ROLAP (Relational OLAP)

* Uses a relational database (SQL-based) instead of prebuilt cubes.
* More scalable for large datasets but slower than MOLAP.

3) HOLAP (Hybrid OLAP)

* Combines the strengths of MOLAP and ROLAP
* Frequency accessed data is stored in cubes (MOLAP), while large datasets remain in relational databases (ROLAP)
* Balances queries performance and storage efficiently.

Advantages

* Fast query performance due to precomputed aggregatio
* Multi dimensional data Analysis for deep insights.
* Enhanced decision-making through interactive reports.
* Scalability to handle large volumes of data

Disadvantages

* High storage requirements for pre-aggregated data.
* Complex implementation and maintanence.
* Latency in data updates. as OLAP is designed for batch processing, not real-time Analytics.

3) Executive Support System

An executive support system (ESS) is a specialized information system designed to help senior executives and top level managers in strategic decision-making. It provides real time, high level data summaries, dashboards and analytical tools to assist in long term planning and performance evaluation. ESS integrates data from multiple sources such as enterprise Resource planning (ERP). Online analytical processing (OLAP) and Business Intelligence (BI) Systems, to provides a holistic view of the organisation.

Architecture of an ESS

1) Data Source Layer.

* Gathers data from internal databases (ERP, CRM, HRM, SCM) and external sources (financial reports, market trends, government statistics)

2) Data processing & Analytics Layer:

* Uses AI driven analytics, OLAP and machine learning to process and summarize data.
* Provides senario analysis, risk assessment, and forecasting tools.

3) Presentation layer

* Displays information in dashboards, reports, charts and KPI summaries.
* Enables executives to interact with data using touch screens, voice commands, or mobile apps.

Key features of an ESS

* High level data summarization.
* Real time data access
* Graphical & interactive UI

External and Internal data integration.

Functions

* Strategic planning.

It helps executives set long-term business goals based on market trends and financial analysis.

* Performance Monitoring

Tracks company performance using KPIs and dashboards.

* Risk management

Identifies potential risks using prediction models and external data.

* Scenario Analysis.

Allows executives to test different business scenarios before making decisions.

* Competitor Analysis

Compares company performance with competitors. using industry benchmarks.

Example of ESS:

* Financial descision making
* Business Expansion strategy.
* Crisis management.